

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is a light green color. They are positioned diagonally, with the blue one in front of the green one.

Modelling NFL Ticket Sales

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Introduction and Investigation Overview

Background: The NFL generates several billions of dollars in revenue annually, making it one of the largest and most valuable organizations in the world. While this income is generated from several sources such as merchandise sales, advertisements and sponsorships, a major portion of it comes from ticket sales. The cost of attending NFL games has significantly increased with the growing popularity of the league, a single trip likely costing a fan at least \$100.

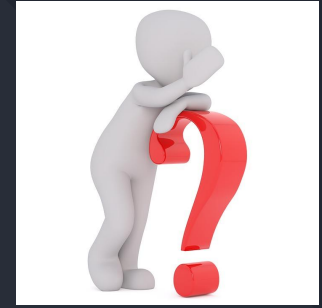
Problem: What variable affects ticket sales in the NFL the most?

Elaboration: Are ticket sales affected more significantly by how many games a team wins, how successful their offense or defense is, etc.? Is there some combination of variables which better explains the amount of tickets sold?

Big Question: Are ticket sales exclusively determined by ticket prices



Big Question: Are ticket sales almost exclusively determined by their price, or are on-field factors significant in predicting sales?



Literature Review



- Purdue University:
 - Study on modelling ticket prices
 - Included variables such as Win ratio, Seat Quality, Team Quality, etc.
- Claremont Colleges:
 - Analysis on the demand for American sports leagues
 - Studied the true willingness of consumers to pay for tickets
 - Used variables to quantify team quality
- Economics of the National Football League
 - Studied the importance of in-person attendance at games despite the ever-growing popularity of the NFL on television
 - Analyzed consumer psychology of purchasing tickets versus watching games remotely



Datasources



Statista: 2010 - 2019 Ticket Sales and Prices For All 32 Teams

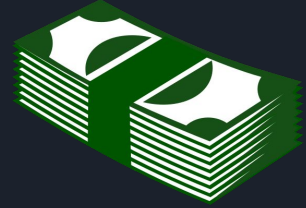
Pro-Football-Reference: Independent variables, namely Wins and SRS statistics

Notes:

- 2020 and 2021 were excluded from the study. This is because the effects of the coronavirus on the NFL would dramatically reduce the usefulness of the model and likely require very complex adjustments. The situation also had little to no precedence, so such a model probably would not be very helpful in making predictions for future ticket sales.
- The study will only include data from the regular season, both for simplicity and because many teams have a very small sample size of data for post season games.



Explanation of Variables



Dependent variable

y : Total revenue from ticket sales, expressed in millions of dollars, for each NFL season from 2010 through 2019. Labelled as “ticketsales” in our regression model

Independent Variables

x_1 : Average ticket price in dollars. Labelled as “avgtickprice”

x_2 : Team wins in previous season. Labelled as “winsprevyear”

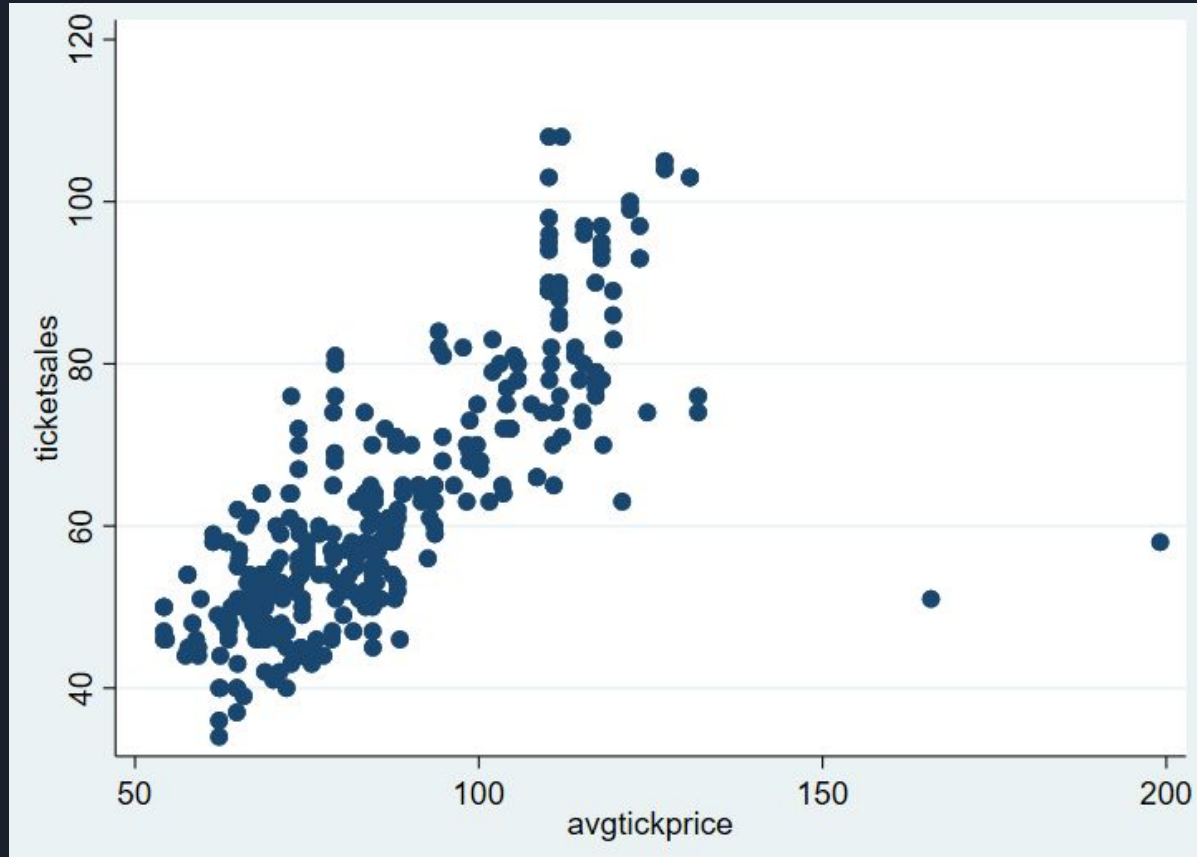
x_3 : Average team wins over previous 5 seasons. Labelled as “avgwinsprev5year”

x_4 : Offensive simple rating system. Labelled as “osrs”

x_5 : Defensive simple rating system. Labelled as “dsrs”

x_6 : Indoor/outdoor stadium. Indicator variable. Labelled as “indoorstadium”

Ticket Sales vs. Ticket Price Scatter Plot



Econometric Model



General Form: $y = \beta_1 + \beta_2 x_1 + \beta_3 x_2 + \beta_4 x_3 + \beta_5 x_4 + \beta_6 x_5 + \beta_7 x_6 + e$

Named Form: $\text{SALES} = \beta_1 + \beta_2 \text{PRICE} + \beta_3 \text{WINS} + \beta_4 \text{FIVEYRWINS} + \beta_5 \text{OSRS} + \beta_6 \text{DSRS} + \beta_7 \text{INDOOR} + e$

Variable Summaries

```
. sum avgtickprice winsprevyear avgwinsprev5year osrs dsrs indoorstadium
```

Variable	Obs	Mean	Std. dev.	Min	Max
avgtickprice	320	86.61297	20.56406	54.2	199.1
winsprevyear	320	7.959375	3.087274	0	15
avgwinspre~r	320	8.00375	2.028137	3	13
osrs	320	.0153125	4.312516	-10.3	14.1
dsrs	320	.01875	3.477229	-9.1	8.9
indoorstad~m	320	.2625	.440682	0	1

Predictions



- Ticket prices: $\beta_2 > 0$
- Wins in previous years: $\beta_3 > 0$
- Average wins over past 5 years: $\beta_4 > 0$
- Offensive Simple Rating System (OSRS): $\beta_5 > 0$
- Defensive Simple Rating System (DSRS): $\beta_6 \approx 0$
- Indoor/Outdoor stadium: $\beta_7 > 0$

Results

```
. reg ticketsales avgtickprice winsprevyear avgwinsprev5year osrs dsrs indoorstadium
```

Source	SS	df	MS	Number of obs	=	320
Model	46667.9054	6	7777.98423	F(6, 313)	=	78.36
Residual	31067.4665	313	99.2570815	Prob > F	=	0.0000
				R-squared	=	0.6003
				Adj R-squared	=	0.5927
Total	77735.3719	319	243.684551	Root MSE	=	9.9628

ticketsales	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
avgtickprice	.5416807	.0292286	18.53	0.000	.4841714	.59919
winsprevyear	-.1447388	.2360275	-0.61	0.540	-.6091399	.3196624
avgwinsprev5year	1.092027	.3668862	2.98	0.003	.3701521	1.813902
osrs	.1544537	.1408645	1.10	0.274	-.1227074	.4316148
dsrs	-.10748	.1681459	-0.64	0.523	-.4383192	.2233592
indoorstadium	-1.063863	1.276777	-0.83	0.405	-3.576014	1.448289
_cons	8.364681	2.966624	2.82	0.005	2.527634	14.20173



Interpretation

Estimated model: $\text{SALES} = 8.36 + 0.54 \text{ PRICE} - 0.14 \text{ WINS} + 1.09 \text{ FIVEYRWINS} + 0.15 \text{ OSRS} - 0.12 \text{ DSRS} - 1.06 \text{ INDOOR}$

- Only average ticket price and the team's average wins over the previous five seasons are significant at $\alpha = 0.05$
- R^2 value of 0.6002 means approximately 60% of the variance in ticket revenue is explained by the model.
- The 95% confidence intervals for the parameters associated with previous year wins, OSRS, and DSRS include 0, meaning they may not be significant in predicting ticket revenue whatsoever.
- Previous year wins, DSRS, and indoor stadium had negative slope coefficients suggesting that as the variables increased, ticket sales decreased.
- Constant coefficient = 8.36



Hypothesis Tests

$H_0: \beta_3 = \beta_5 = \beta_6 = \beta_7 = 0$ vs. H_1 : At least one inequality. $\alpha = 0.05$

Test Statistic: $\text{num} = ((SSR_f - SSR_r)/(p-g)) / (SSE_u/(n-k)) = 0.23$

Rejection Region / Critical Value: $F_{0.05, 4, 313} = 78.36$

Conclusion: Using $\alpha = 0.05$, we *do not* reject the null hypothesis because the test statistic is *less* than the critical value. In the context of our investigation, this means that wins in the previous year, OSRS, DSRS, and indoor stadiums are *not* useful in predicting ticket sales, suggesting the reduced model is more useful



Point Estimation

- A team has won exactly 8 games in each of the previous 5 seasons, plays its home games outdoors, has an OSRS of 0, a DSRS of 0, and their average ticket price is \$80. How much would we expect them to earn in ticket sales for the current season?
- Point Estimate: $x_1 = 80$, $x_2 = 8$, $x_3 = 8$, $x_4 = 0$, $x_5 = 0$, $x_6 = 0$

Sales =

$$8.364 + 0.541(80) - 0.145(8) + 1.092(8) + 0.154(0) - 0.107(0) - 1.064(0)$$

$$= 59.22 \rightarrow 59.22 \text{ million dollars}$$

Interval Estimation

- Interval Estimate: 95% confidence interval

$$b_1 + b_2(80) + b_3(8) + b_4(8) + b_5(0) + b_6(0) + b_7(0) \pm t_{0.05/2, 320-7}(\text{se}(b_1 + 80b_2 + 8b_3 + 8b_4))$$

$$= 59.22 \pm 1.96(11.279)$$

$$= (37.113, 81.327)$$

. vce

Covariance matrix of coefficients of regress model

	e(V)	avgtickp~e	winsprev~r	avgwinsp~r	osrs	dsrs	indoorst~m	_cons
avgtickprice	.00085431							
winsprevyear	-.00018675	.05570898						
avgwinsp~r	-.00282394	-.04685823	.1346055					
osrs	-.00017955	-.00408259	-.00810789	.01984281				
dsrs	-.00021353	-.00715763	.0042759	-.0040495	.02827305			
indoorst~m	.00168251	-.01569743	-.01086234	-.0113247	.01663838	1.6301603		
_cons	-.05034065	-.0478751	-.45690176	.11568455	.03640578	-.36190203	8.80086	



Model Analysis

Reduced model: $\text{SALES} = 7.42 + 0.54 \text{ PRICE} + 1.02 \text{ FIVEYRWINS}$

- Strengths
 - Inclusion of several variables
 - Simplicity
- Weaknesses
 - Cannot fully account for randomness in ticket sales
 - Relatively narrow scope
 - Did not include several variables which may have been significant
 - Variables representing the interaction between two or more independent variables
 - Confidence intervals are not particularly precise



Conclusion

- Ticket price is significant in predicting revenue from ticket sales
- The success of a team over previous seasons is somewhat significant in predicting revenue from ticket sales
- There is a great deal of randomness in the amount of revenue generated from ticket sales, which could potentially be better explained by other variables, such as region, or average age or annual income of buyers
- It may be impossible to perfectly predict ticket sales due to the inherent high demand of the tickets and outside factors outside the control of teams



Works Cited and References

-Ticket sales of each NFL team from 2010 to 2019:

<https://www.statista.com/search/?q=nfl+ticket+sales+2010-2020&Search=&qKat=search&newSearch=true>

-Average ticket price for each NFL team from 2010 to 2019:

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-Source of independent variables: <https://www.pro-football-reference.com/teams/rav/index.htm>

-Literature review sources:

-<https://docs.lib.purdue.edu/cgi/viewcontent.cgi?article=1482&context=jpur>

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-https://www.researchgate.net/publication/302501931_Economics_of_NFL_attendance

-Stata used for linear regression and hypothesis testing

-Excel used for data organization and calculation